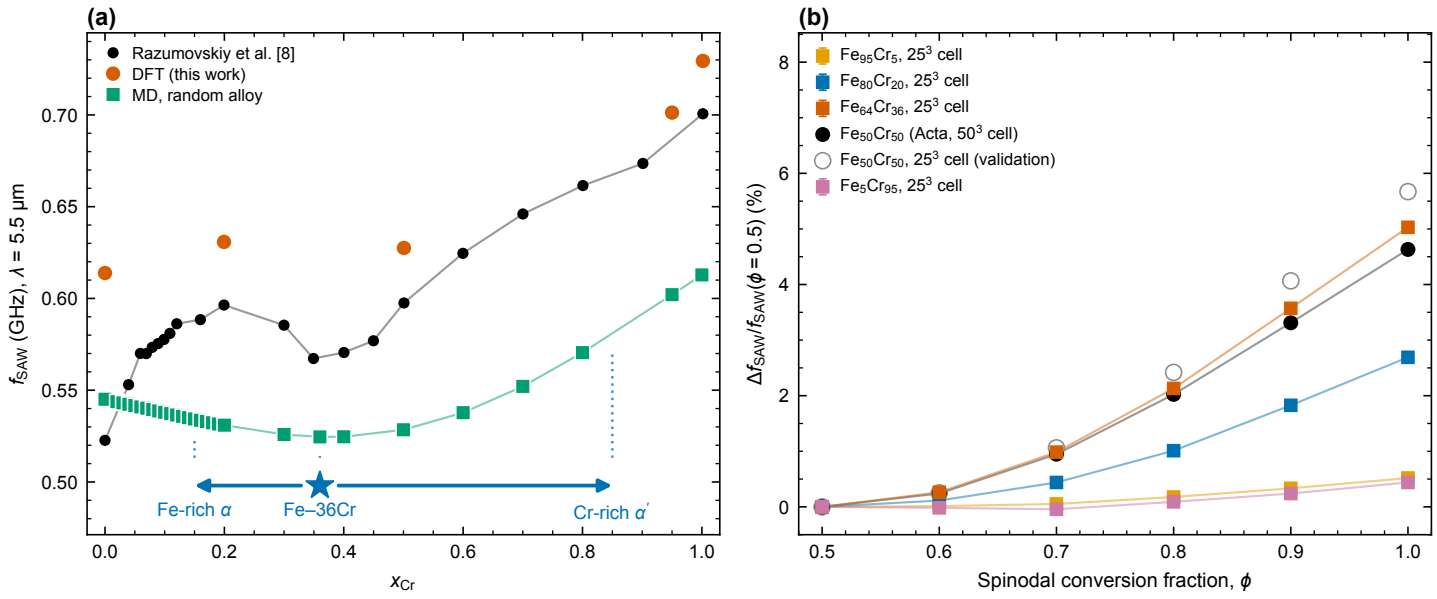
(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
f_{SAW} (GHz), λ = 5.5 μm
0.50
0.55
0.60
0.65
0.70
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}
Fe-rich α
Fe–36Cr
Cr-rich α′
(b)
$Fe_{95}Cr_5$, 25^3 cell
$Fe_{80}Cr_{20}$, 25^3 cell
$Fe_{64}Cr_{36}$, 25^3 cell
$Fe_{50}Cr_{50}$ (Acta, 50^3 cell)
$Fe_{50}Cr_{50}$, 25^3 cell (validation)
Fe_5Cr_{95}, 25^3 cell
$\Delta f_{SAW}/f_{SAW}(\phi = 0.5)$ (%)
0
2
4
6
8
0.5
0.6
0.7
0.8
0.9
1.0
Spinodal conversion fraction, ϕ